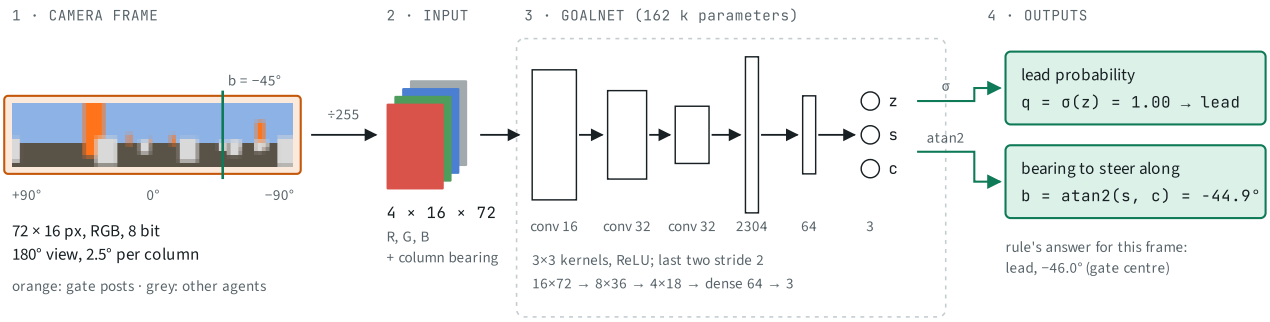
SWARM GATE PASSING · PROGRESS NOTE

Camera-based gate perception for the Ferrante et al. flocking model

Each agent gets a low-resolution forward camera. A small convolutional network (GoalNet) turns one camera frame into two things: whether the agent should lead (a gate is ahead) and the bearing it should steer along. That bearing becomes the leader's goal direction in the Ferrante et al. virtual-force model, which is otherwise unchanged.



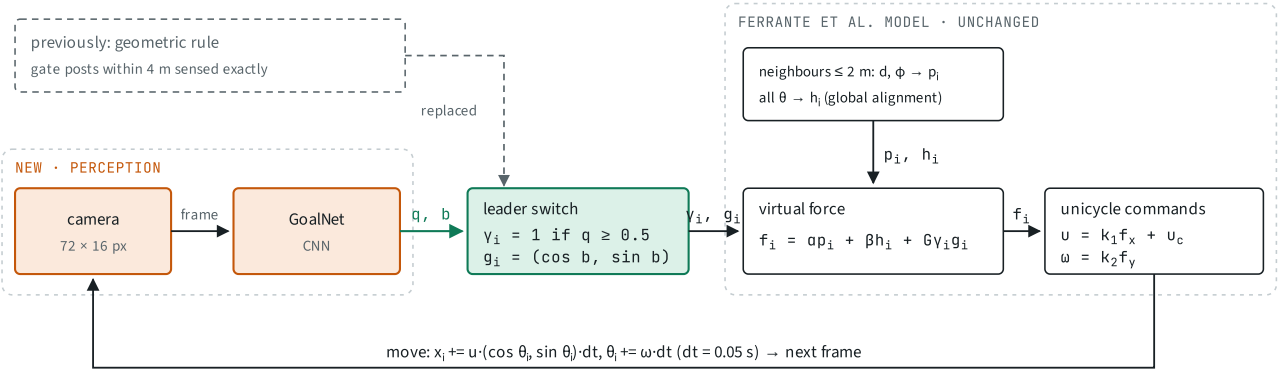
One frame from an agent inside the swarm, 55 s into a run on the default three-gate slalom. The tall bar left of centre is the near post of gate 1; the small one far right is the other post. The network leads and aims between them, about 1° from the rule it was trained to imitate. The fourth input channel holds each column's bearing, so positions in the image map directly to angles.

Inputs

- One frame from the agent's own camera, every simulation step (0.05 s). No memory between frames.
- Rendered in a flat world: posts are 1 m tall orange cylinders, agents 0.25 m grey cylinders. Nearer objects hide farther ones; far posts fade into partial pixels, so distance is only visible as apparent size.
- Nothing else: no position, heading, speed, neighbour data or gate index.

Outputs

- $q \in [0, 1]$: probability that a gate target is ahead. At $q \geq 0.5$ the agent becomes a gate leader.
- b: bearing relative to the agent's heading, positive to the left. Predicted as (sin, cos) to avoid the ±180° wrap.
- Body frame: (cos b, sin b) is the model's goal vector g_i directly.



The per-agent loop, run for all 25 agents every step. Camera and network (orange) take the place of the geometric rule (dashed). The leader switch (green) turns the network output into the same two quantities the rule produced: whether the agent leads (γ_i) and its goal vector (g_i). Everything from the virtual force onward is the Ferrante et al. model as before. The agent's motion changes what its camera sees next, so the network is evaluated in closed loop.

Embedding in the model

Per agent i and step, with the network's outputs q, b and the agent's heading θ_i:

$\gamma_i = 1$ if $q_i \geq 0.5$, else 0	leader or follower
$\psi_i = \theta_i + b_i$	goal heading, world frame
$g_i = (\cos(\psi_i - \theta_i), \sin(\psi_i - \theta_i)) = (\cos b_i, \sin b_i)$	goal vector, body frame
$f_i = \alpha \cdot p_i + \beta \cdot h_i + G \cdot \gamma_i \cdot g_i$	virtual force, body frame
$v_i = \text{clip}(k_1 \cdot f_x + v_c, 0, v_{max})$	forward speed
$\omega_i = \text{clip}(k_2 \cdot f_y, -\omega_{max}, \omega_{max})$	turn rate

Green lines are the network's contribution; the rest is unchanged. The stochastic, stability-based promotion of random-walk leaders is switched off in these runs, so gate leaders are the only leaders.

Training

- **Labels** come from the geometric rule with perfect sensing: lead if a gate post is within 4 m in front; aim at the gate centre if both posts are sensed, else at the one post.
- **Behaviour cloning:** the rule drives 40 runs on random slalom and winding courses; 172 k labelled frames.
- **DAgger:** 3 rounds of 20 runs in which the network drives and the rule labels what it sees; 438 k frames in total. About 40 min on 22 CPU cores.
- **Loss:** binary cross-entropy on lead, plus 4 × squared error on (sin, cos) where the rule leads.

Why the full rule as teacher

The first teacher was the rule restricted to posts the camera shows. It fails: only 1 of 10 slalom runs got most of the swarm through. An agent between the posts sees only the post in front and turns into it, because the rule's one-post case assumes it would also sense the post behind. Trained on the full rule, the network instead learns from the image that a lone post close beside it means it is passing through.

Results

Offline, on held-out frames: lead decision matches the rule on 98.8%; bearing error median 1.8°, 90th percentile 5.3° (9.7° before DAgger). Closed loop, 25 agents, swarm starting toward the course:

Courses	Gate perception	Agents through each gate	Runs with > 50% through all
Default slalom, 3 gates, 10 seeds	Rule (perfect sensing)	92 / 90 / 81%	10 / 10
	GoalNet (camera)	90 / 91 / 71%	10 / 10
Winding, 5 gates 7 m apart, 20 seeds	Rule (perfect sensing)	93 / 57 / 40 / 31 / 25%	5 / 20
	GoalNet (camera)	91 / 61 / 40 / 28 / 24%	4 / 20

The network drives about as well as the rule it imitates. On winding courses both lose the swarm at sharp turns (75° and more), where the next gate is out of sensing range.

Open points

- Single frames: a post that leaves the view is forgotten. Frame stacking or a recurrent layer would add memory.
- Easy visuals: posts are the only orange objects. Textures, lighting and noise would make the task realistic.
- DAgger's closed-loop benefit is not yet measured separately from behaviour cloning; the lead threshold of 0.5 is untuned.